



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Subject: Programming with Java

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ASSIGNMENT No: 5

Title

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Outcome

ArrayList was successfully used to dynamically store and manage a changing list of items, allowing tasks and courses to be added, removed, and viewed at runtime without fixing the size in advance, while StringBuffer was used to efficiently build and display the list contents as a single, readable block of text.

Conclusion

This experiment showed how ArrayList provides a flexible, resizable alternative to plain arrays, making it easy to add or remove elements as data changes during program execution. StringBuffer complemented this by allowing text to be appended and modified efficiently, without repeatedly creating new String objects. Vector works in a similar way to ArrayList but synchronizes its methods, making it a safer choice in multi-threaded programs. Building the To-Do List and Course Registration applications demonstrated practical, real-world use cases of dynamic data storage and string manipulation, which are essential for writing efficient, maintainable Java programs.

Exercise

1. To-Do List Application

An ArrayList stores the tasks so items can be added and removed freely, and a StringBuffer neatly builds the final list for display.

```
1  import java.util.ArrayList;
2  public class ToDoList {
    Run | Debug
3      public static void main(String[] args) {
4          ArrayList<String> tasks = new ArrayList<>();
5          tasks.add(e: "Take out the trash");
6          tasks.add(e: "Complete Java assignment 5");
7          tasks.add(e: "Take doctor's appointment for Aman C");
8          tasks.add(e: "Go cycling");
9
10         tasks.remove(o: "Complete Java assignment 5");
11
12         // StringBuffer builds the final text efficiently
13         StringBuffer taskList = new StringBuffer();
14         taskList.append(str: "---- My To-Do List ----\n");
15         for (int i = 0; i < tasks.size(); i++) {
16             taskList.append((i + 1) + ". " + tasks.get(i) + "\n");
17         }
18         System.out.println(taskList);
19     }
20 }
```

Output:

```
---- My To-Do List ----
1. Take out the trash
2. Take doctor's appointment for Aman C
3. Go cycling
```

2. Student Course Registration System

An ArrayList stores the courses a student has registered for, with methods to add, remove, and view courses. StringBuffer is used to generate a clean, formatted list of registered courses.

```

1  import java.util.ArrayList;
2  public class CourseRegistration {
3      static ArrayList<String> registeredCourses = new ArrayList<>();
4      static void addCourse(String course) {
5          registeredCourses.add(course);
6          System.out.println(course + " added successfully.");
7      }
8      static void removeCourse(String course) {
9          if (registeredCourses.remove(course)) {
10             System.out.println(course + " removed successfully.");
11         } else {
12             System.out.println(course + " not found in the list.");
13         }
14     }
15     static void viewCourses() {
16         StringBuffer courseList = new StringBuffer();
17         courseList.append(str: "---- Registered Courses ----\n");
18         for (int i = 0; i < registeredCourses.size(); i++) {
19             courseList.append((i + 1) + ". " + registeredCourses.get(i) + "\n");
20         }
21         System.out.println(courseList);
22     }
23     Run | Debug
24     public static void main(String[] args) {
25         addCourse(course: "Data Structures");
26         addCourse(course: "Java Programming");
27         addCourse(course: "Critical Thinking");
28         viewCourses();
29         removeCourse(course: "Critical Thinking");
30         viewCourses();
31     }

```

Output:

```

Data Structures added successfully.
Java Programming added successfully.
Critical Thinking added successfully.
---- Registered Courses ----
1. Data Structures
2. Java Programming
3. Critical Thinking

Critical Thinking removed successfully.
---- Registered Courses ----
1. Data Structures
2. Java Programming

```